Continuous Convergence and Curried Functional Spaces

1. The language of filters

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter
$$F = \{A \subset X \,|\, x_n \in A \text{ for all but finitely many } n\}$$
converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

2. Curried functions

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The set $H_n(\mathbb{R})$ of *curried n-variable functions* is a topological vector space defined as follows:

- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;
- If $H_n(\mathbb{R})$ is defined, we define $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise topological and linear structure. In other words, a filter $F \in \mathcal{F}(H_{n+1}(\mathbb{R}))$ converges to $f \in H_{n+1}(\mathbb{R})$ if and only if $F(x) = \{A(x) \mid A \in F\}$ converges to $f(x)$ for all $x \in \mathbb{R}$. It is not hard to see that this topology is compatible with the pointwise vector structure.

Definition 2.2. Let $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ be a function. Define
$$\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, x_2, ..., x_n).$$

Then $\gamma : \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma(\tilde{f})$ will be called the *curried version* of $\tilde{f}$.

3. Continuous functions

Definition 3.1. (compact-open topology): Let X and Y be two topological spaces, and by $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y. For each compact set $K \subset X$ and each open set $U \subset Y$, let $V(K, U)$ be the set of all functions $f \in \mathcal{C}(X, Y)$ such that $f(K) \subset U$. The collection

$$\{V(K, U) \mid K \subset X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{C}(X, Y)$, called the *compact-open* topology.

Definition 3.2. One may try to decouple the notion of convergent filters from topology. Let X be a set with a relation $\to \subset \mathcal{F}(X) \times X$. Then $(X, \to)$ is called a *convergence space* if

- For all $x \in X$, we have $[x] \to x$;
- If $F \to x$ and $G \to x$, then $F \cap G \to x$;
- If $F \to x$ and $F \subset G$, then $G \to x$.

Definition 3.3. A map $f : X \to Y$ between convergence spaces is said to be *continuous* if $f(F) \to f(x) \in Y$ whenever $F \to x \in X$.

Definition 3.4. Let X, Y be two convergence spaces. Then the product $X \times Y$ is endowed with the following convergence structure: a filter $F \in \mathcal{F}(X \times Y)$ converges to $(x, y) \in X \times Y$ if and only if $p_X(F) \to x \in X$ and $p_Y(F) \to y \in Y$.

Definition 3.5. (see [2], pages 25-26): Let X, Y be convergence spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y. Then $\mathcal{C}(X, Y)$ can be endowed with a convergence structure as follows. For a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$, we say that $F \to f \in \mathcal{C}(X, Y)$ if and only if $F(P) \to f(p)$ whenever $P \to p \in X$. Here, the filter $F(P)$ is based on

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The resulting convergence structure on $\mathcal{C}(X, Y)$ is called the structure of *continuous convergence.*

Remark 3.1. Continuous convergence on $\mathcal{C}(X, Y)$ is the coarsest convergence structure that makes the *evaluation mapping*

$$\omega : \mathcal{C}(X, Y) \times X \to Y$$

continuous. Here, the product $\mathcal{C}(X, Y) \times X$ is endowed with the product convergence structure as per Definition 3.4.

Proposition 3.1. (see [2], page 27): *Let X, Y, Z be convergence spaces. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y, Z))$$

defined as $\gamma(f)(x)(y) = f(x, y)$, is well-defined and is a homeomorphism.

Proposition 3.2. (see [2], page 34): *Let X, Y be topological spaces such that X is locally compact and Y is regular. Then the convergence space $\mathcal{C}(X, Y)$ is topologized by the compact-open topology (see [3]). That is, the convergence relation arising from this topology coincides precisely with the convergence structure given in Definition 3.5.*

Proposition 3.3. (see [2], page 28): *Let X be a convergence space and let Y be a Hausdorff convergence space (i.e. every filter in Y converges to at most one point). Then $\mathcal{C}(X, Y)$ is also a Hausdorff convergence space.*

Proposition 3.4. (see [2], page 29): *Let X be a convergence space and Y be a convergence vector space, i.e. a set endowed with both structures such that the operations $+ : Y \times Y \to Y$ and $\cdot : \mathbb{R} \times Y \to Y$ are continuous. Then $\mathcal{C}(X, Y)$ is also a convergence vector space.*

Corollary 3.1. *If X is a locally compact topological space and Y is a regular topological vector space, then $\mathcal{C}(X, Y)$ is a topological vector space.*

Proposition 3.5. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k\in\mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ (in the sense that its derived filter converges) if and only if $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x$ in X.*

Proof: First, let F be the derived filter of $\{f_k\}_{k\in\mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k\in\mathbb{N}}$ converging to $x \in X$, with its derived filter P. Now let V be a neighborhood of $f(x)$ in Y. By the definition of continuous convergence, we have $F(P) \to f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0,m_0} = \{f_n(x_m) \,|\, n \geq n_0,\ m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. First let us show that for any subsequence $\left\{f_{n_k}\right\}_{k\in\mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k\to\infty]{} f(x)$.
To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n\in\mathbb{N}}$ converges to x. Hence we have $f_k(z_k) \xrightarrow[k\to\infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}\left(z_{n_k}\right) \xrightarrow[k\to\infty]{} f(x). \tag{1}$$

Now, to show that $f_k \to f$ in $\mathcal{C}(X, Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f. To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m\in\mathbb{N}}$ of neighborhoods of x. To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_n}(x_m) \notin V$.
In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P, we see that $x_n \xrightarrow[n\to\infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n\to\infty]{} f(x)$ by (1), a contradiction. ∎

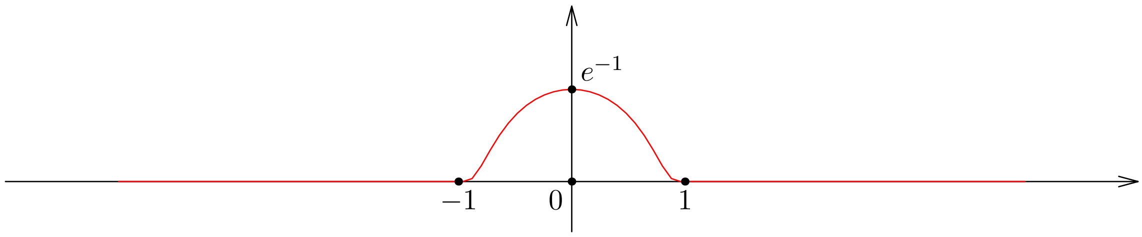
Definition 3.6. Let $n \in \mathbb{N}_0$. The set $C_n(\mathbb{R})$ of *curried continuous functions* is a Hausdorff topological vector space defined recursively as follows:

- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$. Proposition 3.2, Proposition 3.3, and Corollary 3.1 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Example 3.1. The topology on $C_n(\mathbb{R})$ does not coincide with the topology induced from $H_n(\mathbb{R})$, unless $n = 0$. To see this, take $n = 1$ and consider the classical function

$$\varphi(x) = \begin{cases} \exp\left(\frac{1}{|x|^2-1}\right), & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

This is a $C^\infty(\mathbb{R})$ function supported on $[-1, 1]$. Its graph looks as follows:



Now consider the following sequence of functions:

$$f_k(x) = e \cdot \varphi\left(k \cdot \left(x - \frac{1}{k}\right)\right).$$

Clearly, f_k is infinitely differentiable on $\mathbb{R}$, supported on $[0, 2/k]$, and has a maximum at $x = 1/k$ with $f_k(1/k) = 1$. It is clear that for all $x \in \mathbb{R}$, we have $f_k(x) \xrightarrow[k\to\infty]{} 0$, so $f_k \to 0$ in the topology induced from $H_1(\mathbb{R})$. However, f_k do not converge to 0 in the native topology of $C_1(\mathbb{R})$. To see this, take $x_k = 1/k$ and observe that $f_k(x_k) \equiv 1 \neq 0$.

A similar argument applies for any other $n \geq 2$. Hence we conclude that $C_n(\mathbb{R})$ is not a subspace of $H_n(\mathbb{R})$.

Proposition 3.6. *Let $n \in \mathbb{N}_0$. Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma(f)(x_1)(x_2)...(x_n) = f(x_1, ..., x_n)$, is well-defined and is a homeomorphism.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 3.1, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ∎

4. Differentiable functions

Definition 4.1. Let $n \in \mathbb{N}$. A curried function $f \in H_n(\mathbb{R})$ is called *differentiable* if for all $x_0 \in \mathbb{R}$ we have

- $f(x_0) \in H_{n-1}(\mathbb{R})$ is differentiable, if $n \geq 2$;
- There exists a continuous function $f'(x_0) \in C_{n-1}(\mathbb{R})$ such that

$$f(x_0 + h) = f(x_0) + h \cdot f'(x_0) + r(h),$$

where $r(h) \in C_{n-1}(\mathbb{R})$ is a function such that $r(h)/h \to 0$ in $C_{n-1}(\mathbb{R})$ as $h \to 0$. Equivalently, we may write

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \in C_{n-1}(\mathbb{R}).$$

Proposition 4.1. *Let $n \in \mathbb{N}$. Then a curried function $f \in H_n(\mathbb{R})$ is differentiable if and only if its counterpart $\tilde{f} = \gamma^{-1}(f) : \mathbb{R}^n \to \mathbb{R}$ is differentiable everywhere on $\mathbb{R}^n$.*

Proof: First, consider a differentiable function $f \in H_n(\mathbb{R})$. We aim to show that $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ is differentiable everywhere. Fix a point $x = (x_1, x_2, ..., x_n) \in \mathbb{R}^n$ and consider a vector $h = (h_1, h_2, ..., h_n) \in \mathbb{R}^n$. By the differentiability of f, $f(x_1)$, $f(x_1)(x_2)$ and so on, we may repeatedly expand

$$\begin{aligned} f(x_1 + h_1) &= f(x_1) + h_1 \cdot f'(x_1) + r_1(h_1), \\ f(x_1 + h_1)(x_2 + h_2) &= f(x_1)(x_2 + h_2) + h_1 \cdot f'(x_1)(x_2 + h_2) + r_1(h_1)(x_2 + h_2) \\ &= f(x_1)(x_2) \\ &\quad + h_1 \cdot f'(x_1)(x_2 + h_2) + h_2 \cdot f(x_1)'(x_2) \\ &\quad + r_1(h_1)(x_2 + h_2) + r_2(h_2), \end{aligned}$$

where $r_1(h_1) \in C_{n-1}(\mathbb{R})$ and $r_2(h_2) \in C_{n-2}(\mathbb{R})$. We continue the expansion until we obtain

$$\begin{aligned} & f(x_1 + h_1)(x_2 + h_2)...(x_n + h_n) \\ & = f(x_1)(x_2)...(x_n) \\ & \left.\begin{aligned} & + h_1 \cdot f'(x_1)(x_2 + h_2)...(x_n + h_n) \\ & + h_2 \cdot f(x_1)'(x_2)(x_3 + h_3)...(x_n + h_n) \\ & + h_3 \cdot f(x_1)(x_2)'(x_3)(x_4 + h_4)...(x_n + h_n) \\ & \vdots \\ & + h_n \cdot f(x_1)(x_2)...(x_{n-1})'(x_n) \end{aligned}\right] (A) \\ & \left.\begin{aligned} & + r_1(h_1)(x_2 + h_2)(x_3 + h_3)...(x_n + h_n) \\ & + r_2(h_2)(x_3 + h_3)...(x_n + h_n) \\ & \vdots \\ & + r_{n-1}(h_{n-1})(x_n + h_n) \\ & + r_n(h_n). \end{aligned}\right] (B) \end{aligned}$$

Observe that all functions $f'(x_1)$, $f(x_1)'(x_2)$, $f(x_1)(x_2)'(x_3)$ that occur in (A), are continuous by the definition of differentiability, and so when $h \to 0$, we have for instance

$$f'(x_1)(x_2 + h_2)...(x_n - h_n) - f'(x_1)(x_2)...(x_n) \to 0.$$

This implies that

$$
\begin{aligned}
&h_1 \cdot (f'(x_1)(x_2+h_2)...(x_n-h_n) - f'(x_1)(x_2)...(x_n)) = o(|h|),\\
&h_2 \cdot \left(f(x_1)'(x_2)(x_3+h_3)...(x_n-h_n) - f(x_1)'(x_2)...(x_n)\right) = o(|h|),\\
&\vdots\\
&h_n \cdot \left(f(x_1)...(x_{n-1})'(x_n) - f(x_1)...(x_{n-1})'(x_n)\right) = o(|h|).
\end{aligned}
$$

Now, consider the convergence $r_1(h_1)/h_1 \xrightarrow[h_1\to 0]{} 0$. This convergence occurs in the space $C_{n-1}(\mathbb{R})$, which means that as $h \to 0$, we have $h_i \to 0$ for all $1 \le i \le n$ and

$$
\left|\frac{r_1(h_1)(x_2+h_2)...(x_n+h_n)}{|h|}\right| \le \left|\frac{r_1(h_1)}{h_1}(x_2+h_2)...(x_n+h_n)\right| \xrightarrow[h_1,h_2,...,h_n\to 0]{} 0.
$$

In other words, we see that $r_1(h_1)(x_2+h_2)...(x_n+h_n) = o(|h|)$, and similarly all remaining terms in (B) are $o(|h|)$. Finally, combining these observations yields

$$
\begin{aligned}
\tilde{f}(x+h) &= f(x_1+h_1)(x_2+h_2)...(x_n+h_n)\\
&= f(x_1)(x_2)...(x_n)\\
&\quad \left.\begin{aligned}
&+h_1 \cdot f'(x_1)...(x_n)\\
&+h_2 \cdot f(x_1)'(x_2)...(x_n)\\
&+h_3 \cdot f(x_1)(x_2)'(x_3)...(x_n)\\
&\vdots\\
&+h_n \cdot f(x_1)...(x_{n-1})'(x_n)
\end{aligned}\right] L(h)\\
&\quad + o(|h|)\\
&= \tilde{f}(x) + L(h) + o(|h|),
\end{aligned}
$$

where $L : \mathbb{R}^n \to \mathbb{R}$ is linear since it has the form $L(h) = \sum_{k=1}^n c_k h_k$. This allows us to conclude that $\tilde{f}$ is differentiable at x.

To show the reverse implication, we conduct an induction over n. If $n = 1$ and $f \in H_1(\mathbb{R})$, its differentiability is clearly equivalent to the usual definition. Assume that the implication holds for $n-1$, and consider a function $f \in H_n(\mathbb{R})$ such that $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ is differentiable everywhere on $\mathbb{R}^n$.
Let $x_1 \in \mathbb{R}$. First of all, since a restriction of a differentiable function is differentiable, we see that the function

$$
f(x_1) = \gamma\left((x_2,...,x_n) \mapsto \tilde{f}(x_1,x_2,...,x_n)\right) \in H_{n-1}(\mathbb{R})
$$

is differentiable by the induction hypothesis. It then remains to show that there is $f'(x_1) \in C_{n-1}(\mathbb{R})$ and $r_1(h_1) \in C_{n-1}(\mathbb{R})$ for all $h_1 \in \mathbb{R}$ such that $r_1(h_1)/h_1 \xrightarrow[h_1\to 0]{} 0$ and

$$
f(x_1+h_1) = f(x_1) + h_1 \cdot f'(x_1) + r_1(h_1).
$$

Being differentiable, $\tilde{f}$ has partial derivatives everywhere, and so we can define

$$
f'(x_1) = \gamma\left(\frac{\partial \tilde{f}}{\partial x_1}\right)(x_1) \in H_{n-1}(\mathbb{R}).
$$

For variable $y_2, y_3, ..., y_n \in \mathbb{R}$ and $h_1 \in \mathbb{R}$, we have

$$\begin{aligned}\tilde{f}(x_1+h_1, y_2, ..., y_n) &= \tilde{f}(x_1, y_2, ..., y_n) + \left(d_{(x_1, y_2, ..., y_n)}\tilde{f}\right)(h_1, 0, ..., 0) + r(h_1, y_2, ..., y_n) \\ &= f(x_1)(y_2)...(y_n) + h_1 \cdot f'(x_1)(y_2)...(y_n) + r(h_1, y_2, ..., y_n),\end{aligned}$$

where $r(h_1, y_2, ..., y_n) = o(h_1)$. We then define $r_1(h_1) = \gamma(r)(h_1)$. Hence, by abstracting over $y_2, ..., y_n$, we obtain

$$f(x_1 + h_1) = f(x_1) + h_1 \cdot f'(x_1) + r_1(h_1).$$

It remains to show that $f'(x_1)$ and $r_1(h_1)$ are continuous and that $r_1(h_1)/h_1 \to 0$ as $h_1 \to 0$. ■

Proposition 4.2. *If $f \in H_n(\mathbb{R})$ is differentiable, then f is continuous.*

Proof: If f is differentiable, then $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ is differentiable, then $\tilde{f}$ is continuous, then f is continuous. ■

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.

[2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.

[3] R. H. Fox, "On Topologies For Function Spaces," 1945.